

HYDRA SAMO

Brand Identity System · Engineering Documentation · Production Handbook

VERSION

1.0

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v4.0.0 — "One Body, Three Heads"

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![[HYDRA SAMO — Certified Mark](../public/hydra-mark.svg)]

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The definitive reference manual for the HYDRA SAMO Brand Identity System. Supersedes the individual engineering documents for reading purposes; the original documents remain the official source files.

- 1. Introduction
- 2. Brand Philosophy
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- 4. Design Principles
- 5. Logo Evolution
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1. Introduction

1.1 The Brand

HYDRA SAMO is a premium portfolio and creative studio specializing in **Video Editing**, **Motion Design**, and **Voice Over** — three disciplines, one unified body of work. The brand is the creative output of **Bendali Issam Eddine** (Hydra Samo), a solo creative studio operating as the sole approval authority for every branding decision.

Field	Value
---	---
Brand owner	Bendali Issam Eddine (Hydra Samo) — solo creative studio
Repository	hydra-samo/HYDRA-SAMO-LABS
Live URL	https://hydra-samo.github.io/HYDRA-SAMO-LABS/
Copyright	Bendali Issam Eddine (Hydra Samo)
License	Proprietary — all rights reserved
System steward	Lead Brand Systems Engineer (production pipeline)

1.2 Purpose of This Book

This publication compiles the **entire HYDRA SAMO Brand Identity System** — from design brief through certification, production, packaging, and release — into a single coherent reference. It consolidates the complete project history so that:

- No design decision is lost.
- No engineering rationale is lost.
- No governance rule is lost.
- The brand can be reproduced from a single authoritative document.

1.3 How to Read This Book

The book is organized into 36 chapters mirroring the brand lifecycle:

- 1
- Foundation (1–4):** what the brand is and why it exists.
- 2
- Design System (5–14):** logo evolution, geometry, construction, color, typography, iconography, motion.
- 3
- Engineering & Integration (15–18):** website architecture, SVG engineering, component implementation.
- 4
- Production (19–32):** asset pipeline, QA, certification, governance, and every deliverable system.
- 5
- Release & Future (33–36):** versioning, final release, and the appendix.

Every chapter references the authoritative source documents in the repository; where information appeared in multiple documents it has been **merged, not duplicated**.

1.4 Lifecycle Summary

Phase	Source docs	Status
---	---	---
Design specification	DESIGN.md , AGENTS.md	■ Complete
Independent reviews	OUTPUT/01 ... 04	■ Complete
Consensus review	OUTPUT/05	■ Complete
Implementation plan	OUTPUT/06	■ Complete

Phase	Source docs	Status
Changelog	OUTPUT/07	■ Complete
Final certification	OUTPUT/08	■ CERTIFIED
Brand freeze	OUTPUT/09	■ LOCKED
Master asset	OUTPUT/10	■ Complete
Export pipeline	OUTPUT/11	■ Complete
Website assets	OUTPUT/12	■ Complete
Motion system	OUTPUT/13	■ Complete
Social system	OUTPUT/14	■ Complete
Print system	OUTPUT/15	■ Complete
Brand governance	OUTPUT/16	■ Complete
File structure	OUTPUT/17	■ Complete
QA validation	OUTPUT/18	■ Complete
Final release	OUTPUT/19	■ Complete
Asset production	OUTPUT/20 + report	■ Complete
Production QA	OUTPUT/21 + report	■ PASSED
Completion audit	OUTPUT/22 + report	■ Complete
Release packaging	OUTPUT/23 + report	■ Complete
Final sign-off	OUTPUT/24 + report	■ COMPLETE
Pipeline execution	OUTPUT/25 + execution	■ Complete
Complete brand system	OUTPUT/26	■ Complete
Brand book	OUTPUT/27	■ This document

2. Brand Philosophy

2.1 One Body, Three Heads

The HYDRA SAMO identity is built on a single organizing idea: **one body, three heads**. The three disciplines — Video Editing, Motion Design, and Voice Over — are not three separate businesses under one name; they are three specialized capabilities growing from one unified creative body. The identity must therefore read as a *single organism*, never as a collection of disconnected marks.

2.2 The Hidden Hydra Principle

The mark must not immediately appear to be a hydra. At first glance it reads as a clean, engineered, tri-symmetric geometric emblem. Only after a closer look does the viewer realize the symbol subtly forms a three-headed serpent. The discovery must feel **intentional and satisfying** — the icon rewards attention without relying on obvious creature details.

2.3 Aesthetic: Bio-Organic Dark Editorial

The visual language is immersive, high-contrast, dark-mode-first editorial showcase:

- Deep Abyssal canvas.
- Mythic Emerald life-signal accents.
- Glassmorphic surfaces over a film-grain atmosphere.
- Engineered precision that still feels organic — a "hydra discovered inside a futuristic interface."

2.4 Banned Anti-Patterns

The following are **strictly forbidden** in every language, locale, and surface:

Anti-pattern	Examples
---	---
Neon cyan	#00FFCC, #00F0FF, #00E5FF, default neon Tailwind colors
Bright blue gradients	Cyan/blue gradient accents
SaaS eyebrow pills	// SECTION_TITLE mono eyebrows, pill wrappers
Comparison matrices	Rigid corporate "Agency vs Hydra" tables
Numbered badges	01, 02 rigid numbered markers
Geo/location badges	ALGERIA // GLOBAL, "Based in ...", country pins — in any locale

These were codified during the review pipeline and remain active governance (DESIGN.md, AGENTS.md).

2.5 Communication Values

The brand communicates: **Precision, Movement, Creativity, Innovation, Craftsmanship, Intelligence, Trust, and Premium quality** — with the quiet confidence of a lasting identity rather than a transient trend.

3. Creative Vision

3.1 The Design Brief

The master logo design brief defines a single iconic SVG brand mark for HYDRA SAMO. The mark must consist of **only one icon** — no text, no initials, no typography, no slogan, no border, no background, no decorative elements.

It should feel timeless, premium, confident, and instantly recognizable — in the company of brands such as Apple, Nike, Vercel, Linear, Framer, Mercedes-Benz, Nothing, and OpenAI. Not trendy. Not flashy. Not illustrative. Not generic AI art. It must still feel modern twenty years from now.

3.2 Vision Statements

- **A hidden symbol discovered inside a futuristic interface** — an abstract geometric emblem that quietly reveals a three-headed hydra.
- **Simple enough to be recognized instantly**; distinctive enough to stand on its own without text; refined enough to become the lasting visual identity.
- **Engineered, not illustrated** — every angle, intersection, and curve appears mathematically intentional.

3.3 The Concept Is Approved

The concept — three heads, one body, one engine — was **unanimously defended** across all independent reviews. The reviews refined the *perceptibility* of the concept, never the concept itself.

4. Design Principles

4.1 Core Principles

- 1 **Identity before decoration.** The "three heads, one body" read is sacred; any treatment that obscures it fails.
- 2 **Monochrome-first.** The design must be perfect in solid black before color; recoloring must be effortless.
- 3 **Single source of truth.** One certified master; every raster, icon, social asset, print file, and motion asset is derived from it. Re-export only, never re-create.
- 4 **Lighting lives outside the SVG.** All glow, shadow, and reflection are external CSS utilities — never baked into the vector.
- 5 **Restraint over ornament.** Minimal anchor points, clean cubic Béziers, generous negative space, no gradients, no effects, no embedded rasters in the master vector.
- 6 **Motion is opt-in and settled.** Approved effects reveal the mark and settle; the only approved loop is the low-key loading pulse.
- 7 **Accessibility is non-negotiable.** **prefers-reduced-motion**, decorative **aria-hidden**, WCAG AA contrast.
- 8 **Vector discipline.** True SVG, pixel-perfect geometry, crisp at 16×16, fully editable in Figma, Illustrator, and Inkscape.

4.2 What the Mark Must Do

The logo must:

- Work as favicon (16 px), app icon, loading animation, watermark, and social avatar.
- Be monochrome-true (**currentColor**).
- Be motion-ready (stroke drawing, glow, pulse, hover) without losing identity.
- Remain legible from 16 px to billboard scale.

4.3 What the Mark Must Never Do

From the negative prompt: no fantasy hydras, realistic snakes, dragons, esports/gaming logos, mascots, clipart, NFT/crypto styling, shields, crests, crowns, wings, eyes, teeth, tongues, scales, tribal patterns, medieval styling, lettermarks, monograms, typography inside the mark, stock-logo aesthetics — or anything that feels aggressive, generic, or obviously AI-generated.

5. Logo Evolution

5.1 Timeline

Version	Description	Status
---	---	---
V1	Original concept	Superseded
V2	Pre-refinement baseline — 3-fold master head path + pointy-top hexagon core (circumradius 14.5)	Superseded
V4	Certified refinement of V2 — locked serpent-head geometry + R17 hex core, shared optical frame	■ CERTIFIED — permanent

5.2 Review Cycle 2026-08-05 (First Review)

The first independent review cycle scored the V2 baseline:

Category	Reviewer	Score
---	---	---
Creative Direction	Creative Director	63 / 100
Construction	Senior Logo Designer	67 / 100
Brand Recognition	Brand Recognition Specialist	53 / 100
Risk / Gatekeeping	Devil's Advocate	54 / 100
Consensus Overall	All reviewers	59 / 100

Verdict: REFINE.

Major Findings

- 1
- Favicon fails at 16 px (Critical) — three heads + hexagon merge into a blob; mark small and off-center in the tile.
- 2
- Optical frame off (High) — mark occupies y 11.5–69.9 of the 0–100 canvas; ~30% dead space below.
- 3
- Wrong perceptual category (High) — reads rotor / drone / fan / generic tech crest, not a creative hydra.
- 4
- Construction kinks (Medium) — non-tangent joints at the throat notch and lower-right base.
- 5
- Off-brand social asset (High) — OpenGraph image steel-blue and under-sized vs the live emerald mark.
- 6
- Unused legacy colorways (Medium) — teal / violet / warm rasters remained in the repo.
- 7
- Hidden hydra invisible (Medium) — concept correct, perceptibility insufficient.

Strengths Recorded

- Concept: three heads, one body, one engine — unanimously defended.
- Exact regular hexagon core (circumradius 14.5); correct 3-fold rotation construction.
- Exemplary SVG discipline: **currentColor**, **pathLength**, no baked decoration, clean markup.
- Consistent website integration across nav / hero / pre-splash / splash.

5.3 Refinement Plan (Priority 1)

The implementation plan (**OUTPUT/06**) defined launch-blocking changes:

- **P1-1** Fix the optical frame (viewBox re-balance → ≥75% coverage, optical center at center).
- **P1-2** Provide a dedicated favicon that reads at 16 px.

- **P1-3** Make the head read as a head, not a blade (broaden base flare, $\approx 83^\circ \rightarrow \approx 55\text{--}60^\circ$ wedge).
- **P1-4** Repair construction tangency (every joint tangent-continuous, zero visible corners).
- **P1-5** Rebuild the OpenGraph / social preview from the single source.
- **P1-6** Retire the legacy raster logo variants.

Priority 2 covered the outline-variant leak, internal weight rebalance, the fate of **.hydra-mark-spin**, cleaning the component API, and documenting minimum sizes and clear space.

5.4 V4 Implementation (Same Day)

Executed on brand-owner approval, the V4 refinement delivered:

- Head geometry rebuilt from two open Catmull-Rom chains + V-base tucking the core top edges.
- Hexagon core enlarged to circumradius 17; shared optical frame reframed to **11 2.94 78 78**.
- **.hydra-mark-spin** retired from CSS and governance (rotor cue).
- Legacy teal / violet / warm rasters retired from the repository.
- All measured verification gates passed (see Chapter 6, 9, 22).

6. Geometry

6.1 Certified Parameters

The geometry sheet below is the certified record, sourced from **MASTER/hydra-mark_v4_master.svg** (**ASSETS/SOURCE/geometry-sheet.md**):

Parameter	Value
---	---
viewBox	11 2.94 78 78 (square)
Paths	4 (3 heads + hex core)
Origin / rotation center	(50, 50)
Head rotations	0°, 120°, 240° (static rotate(...) only)
Head tip (head 0)	(50, 11.50)
Head → core junction	(50, 33.00)
Head height (safe-zone module)	21.5 u
Core hexagon vertices	(50,33) (64.72,41.5) (64.72,58.5) (50,67) (35.28,58.5) (35.28,41.5)
Core circumradius	17 u
Tip circle radius	38.5 u (through head tips)
Full mark extent	x 16.65...83.35 , y 11.5...72.2 (incl. rotated head tips)
pathLength	1 on all four paths (for draw/trace math)

6.2 The Construction Grid

![V4 Construction Diagram](../ASSETS/SOURCE/hydra-mark_v4_construction.svg)

- **Unit grid:** 5 u — every coordinate on the 5-grid.
- **Construction hex:** pointy-top, circumradius 36, center **(50, 50)** — the invisible hexagonal grid the mark is built on.
- **Tip circle:** radius **38.5** through all three head tips.
- **Core hex:** circumradius **17**, inscribed on the tip circle's inner harmonic.
- **120° spokes:** from the center through each head tip.

6.3 Head Path (Master)

The single master serpent-head path (head 0, pointing up):

```
<font face="Courier" size="7.4" color="#d6e2e8">M 50.00 11.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 50.56 12.05, 52.20 13.55, 53.36 14.80</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 54.52 16.05, 55.68 17.37, 56.96 19.00</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 58.24 20.63, 60.44 22.73, 61.04 24.60</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 61.64 26.47, 60.84 28.40, 60.56 30.20</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 60.28 32.00, 59.53 34.02, 59.36 35.40</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 59.19 36.78, 59.50 37.98, 59.52 38.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 50.00 33.00</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 40.48 38.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 40.50 37.98, 40.77 36.78, 40.64 35.40</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 40.51 34.02, 39.40 31.97, 39.68 30.20</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 39.96 28.43, 41.56 26.40, 42.32 24.80</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 43.08 23.20, 43.52 22.27, 44.24 20.60</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 44.96 18.93, 45.68 16.32, 46.64 14.80</font>
<font face="Courier" size="7.4" color="#d6e2e8">C 47.60 13.28, 49.44 12.05, 50.00 11.50 Z</font>
```

6.4 Core Path (Hexagon)

```
<font face="Courier" size="7.4" color="#d6e2e8">M 50.00 33.00</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 64.72 41.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 64.72 58.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 50.00 67.00</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 35.28 58.50</font>
<font face="Courier" size="7.4" color="#d6e2e8">L 35.28 41.50 Z</font>
```

6.5 Key Measurements

Metric	Value	Note
---	---	---
Wedge / head flare	57.5° per head	Target 55–60° — heads read as serpent hoods, not rotor blades
Tangency	0.0°	Crown / throat / neck joints tangent-clean
Frame coverage	85.5% × 78.0%	Optical center at canvas center
16 px ink coverage	42%	Per-120° lobe ink 31/39/37 — three balanced heads
Core visibility at 16 px	Visible	Body reads, not fan
Size gain vs V2 frame	~40%	Reframed 0 0 100 100 → 11 2.94 78 78

7. Construction

7.1 The Four Paths

The mark is **four paths, never merged**:

- 1 **hydra-head-0** — the master head (up), full path data.
- 2 **hydra-head-1** — the master head rotated **rotate(120 50 50)** (down-left → Motion Design).
- 3 **hydra-head-2** — the master head rotated **rotate(240 50 50)** (down-right → Voice Over).
- 4 **hydra-core** — the R17 pointy-top hexagon (the one body).

No path merging — the three heads stay separate from the hex core so stroke-drawing and per-class styling remain possible.

7.2 Build Order

- 1 Draw the **pointy-top hexagonal grid** (construction hex circumradius 36).
- 2 Inscribe the **head tip circle** (radius 38.5).
- 3 Place the master serpent-head path with its tip at **(50, 11.50)** and its V-base junction at **(50, 33.00)**.
- 4 Replicate in exact 3-fold rotational symmetry about **(50, 50)**.
- 5 Seat the R17 **hexagon core** so the three neck waists flare into it — the heads merge into the body.

7.3 Class Contract

```
<font face="Courier" size="7.4" color="#d6e2e8">&lt;path class="hydra-head hydra-head-0" d="..." pathLength={1} /&gt;</font>
<font face="Courier" size="7.4" color="#d6e2e8">&lt;path class="hydra-head hydra-head-1" d="..." transform="rotate(120
<font face="Courier" size="7.4" color="#d6e2e8">&lt;path class="hydra-head hydra-head-2" d="..." transform="rotate(240
<font face="Courier" size="7.4" color="#d6e2e8">&lt;path class="hydra-core" d="..." pathLength={1} /&gt;</font>
```

The classes are stable and public — CSS and Framer Motion target them for stroke drawing, glow, pulse, and hover without mutating path data.

7.4 Construction Rules

- 1 Never reshape **d**; only rotate/scale the certified paths.
- 2 No text, initials, borders, or gradients inside the mark.
- 3 Glow/reflections live in CSS (**.hydra-mark-glow**), never in the SVG.
- 4 Rotation animation is forbidden (**.hydra-mark-spin** retired).

8. Hexagonal Grid System

8.1 The Invisible Construction Grid

The logo is engineered on a **pointy-top hexagonal grid** — the same system used by the review pipeline to validate angles, intersections, and optical balance. Nothing is random:

- **Hexagonal proportions** — every form relates to the hex.
- **Geometric alignment** — vertices, junctions, and spokes land on grid coordinates.
- **Symmetry** — exact 3-fold rotational symmetry about the optical center.
- **Consistent spacing** — one clear-space module (one head height, 21.5 u).
- **Balanced visual weight** — center-dominated, equally weighted around the center.
- **Controlled negative space** — the deep concave throat notches carve entirely through negative space.

8.2 Grid Hierarchy

Element	Radius / size	Role
---	---	---
Construction hex	circumradius 36	Invisible building grid
Tip circle	radius 38.5	Through the three head tips
Core hex	circumradius 17	The one body — structural anchor
Head module	21.5 u high	Serpent head + clear-space module

8.3 Pointy-Top vs Flat-Top

The core is a **pointy-top** hexagon (vertices at top and bottom), matching the head-up orientation of head 0. V2's core was circumradius 14.5; V4 enlarged it to 17 so the solid core anchors the form and kills the rotor / spinner read while preserving the outer silhouette.

8.4 Weight Distribution

Mass was redistributed from the arm stems into the center (~+20% **optical weight**, center-dominated). Heads are bolder (~13–14 units) with a clear neck constriction (~8 units); the solid core anchors the mark. The outer silhouette retains ≥80% overlap with V1; snout radius and flank extent are unchanged.

9. Optical Corrections

The V4 refinement was driven entirely by measured optical defects from the first review.

9.1 Frame Re-Balance (P1-1)

Before: shared 0 0 100 100 canvas, mark occupying y 11.5–69.9, ~30% dead space below, floating high.

After: shared optical frame `viewBox="11 2.94 78 78"`.

- Frame coverage **85.5% × 78.0%** (≥75% gate).
- Optical center placed at canvas center.
- Mark ~40% larger in every fixed UI slot.

9.2 Favicon at 16 px (P1-2)

The 16 px render previously merged into a blob. Post-refinement:

- Three distinct head masses + a **visible center core**.

- Ink coverage **42%**; per-120° lobe ink 31/39/37 (balanced).
- Optically centered at **85.4% × 78.1%** of the tile (≥75% gate).

9.3 Perceptual Category (P1-3)

Head flare tightened to a **57.5° wedge** (target 55–60°) with bold crown/hood, a deep concave throat notch, and visible neck waists. The mark now reads *serpent heads on one body* — not a rotor, drone, or fan. The forbidden continuous-rotation cue was retired.

9.4 Tangency Repair (P1-4)

Rebuilt head path (two open Catmull-Rom chains + V-base) is **tangent-continuous** — **0.0°** across crown, throat, and neck joints. No corners, no visible angles at any zoom; one continuous serpentine curve.

9.5 OpenGraph Rebuild (P1-5)

`public/hydra_logo.jpg` rebuilt 1024×1024: abyssal **#060c09** field, emerald **#10b981** mark, centered at **84.6% × 77.1%** coverage — the real emerald mark, not the old steel-blue asset.

9.6 Legacy Cleanup (P1-6 / P2)

- Teal / violet / warm logo rasters **retired** from `src/assets/images/` (kept `hydra_samo.webp` portrait).
- `.hydra-mark-spin` **removed** from CSS, keyframes, and reduced-motion references.
- Unused **animated** component prop removed.
- Outline-variant interior leak **fixed** — no interior tail strokes cross the transparent hexagon.

10. Symbol Meaning

10.1 Three Heads, One Body

Each of the three forms is a serpent head in profile — pointed snout, bold crown/hood, and a deep concave throat notch carved entirely through negative space:

Head	Position	Discipline
---	---	---
Head 0 (up)	0°	Video Editing
Head 1 (down-left)	120°	Motion Design
Head 2 (down-right)	240°	Voice Over

Each head overhangs a distinct neck waist that flares back out into the solid hexagon core: **one body, three heads**.

10.2 The Hidden Hydra

At a glance the mark reads as a clean, engineered tri-sym emblem. On closer inspection the hydra appears. The discovery is intentional — the logo rewards attention without relying on obvious creature details.

10.3 The One Body

The hexagon core is a **genuine structural anchor** the three necks flow into — not a hub or dot. It is the "one unified vision" from which all three disciplines grow.

10.4 The Emblem Read

The mark is monochrome-true: it reads as a compact **triangular-hex emblem** even at 16 px, where the internal throat-notch detail naturally dissolves into a clean silhouette.

11. Typography

11.1 Font Stacks

Canonical tokens in `src/index.css@theme`:

Token	Font	Usage
---	---	---
<code>--font-sans</code>	Manrope	EN/FR body (400), UI labels (500), buttons/CTAs (600)
<code>--font-display</code>	Space Grotesk	Wordmark (700), hero (700), primary section titles (600), secondary headings (500)
<code>--font-arabic</code>	IBM Plex Sans Arabic	Arabic headings (600, 500 secondary) and body (400), applied via <code>html.lang-ar</code> overrides
<code>--font-mono</code>	JetBrains Mono	Code and technical readouts only (400, 600 emphasis)

Wired from Google Fonts with deliberately limited weights for performance:

```
<font face="Courier" size="7.4" color="#d6e2e8">SpaceGrotesk:wght@500;600;700</font>
<font face="Courier" size="7.4" color="#d6e2e8">Manrope:wght@400;500;600</font>
<font face="Courier" size="7.4" color="#d6e2e8">IBM+Plex+Sans+Arabic:wght@400;500;600;700</font>
<font face="Courier" size="7.4" color="#d6e2e8">JetBrains+Mono:wght@400;600</font>
```

11.2 Type Hierarchy

Element	Spec
---	---
Logo wordmark	Space Grotesk 700, uppercase; tracking -0.02em (nav / pre-splash), -0.03em (splash title)
Hero display	Space Grotesk 700, text-4xl → lg:text-7xl , tracking-tight , leading-[1.02]
Primary section titles (h2)	Space Grotesk 600, text-3xl → md:text-5xl , uppercase, max 1–2 words in Mythic Emerald
Secondary headings (h3/h4)	Space Grotesk 500

Element	Spec
Body (EN/FR)	Manrope 400, leading-relaxed
UI labels & eyebrows	Manrope 500, text-xs/text-sm , uppercase, tracking-widest — not mono
Buttons & CTAs	Manrope 600, uppercase, tracking-wider
Arabic	IBM Plex Sans Arabic 600 headings / 400 body via html.lang-ar
Code / technical meta	JetBrains Mono — software-stack pills, timestamps, video overlay badges, metric readouts, literal code refs

11.3 Mono = Technical Only

JetBrains Mono is reserved for code and data readouts. Every prose eyebrow, form label, and menu label renders in Manrope 500. The **// SECTION** mono-eyebrow pattern stays banned.

11.4 Responsive Scale

- Fluid step-up per breakpoint — no single-size headings.
- Arabic: **html.lang-ar** scales text steps ~+30% (**.text-xs** → 15px ... **.text-7xl** → 88px) and zeroes letter-spacing — Arabic reads smaller and breaks apart under wide tracking.

11.5 Accessibility

- Contrast-driven palette: off-white **#f3f4f6** titles, muted **#94a3b8** body, emerald **#10b981** accents on abyssal **#060c09** — AA-friendly for large/bold type.
- Interactive text never smaller than **text-xs/text-sm** with **min-h-[44px]** touch targets.
- **prefers-reduced-motion** skips the splash letter-spacing reveal.
- px-based Tailwind utilities scale cleanly under browser zoom (200%).

12. Color System

12.1 Certified Palette

![[HYDRA SAMO Palette]](hydra-brand-palette.svg)

Token	Hex	Role
---	---	---
Mythic Emerald	#10b981	Primary brand accent — default mark color, all surfaces
Deep Jade	#059669	Secondary accent — depth, hovers, secondary detail
Soft Mint	#34d399	Ambient glow / highlight
Abyssal Slate	#060c09	Primary canvas (dark mode)
Bone	#f3f4f6	Typography primary (off-white)
Slate Muted	#94a3b8	Typography muted

12.2 Monochrome Modes

Mode	Color	Use
---	---	---
White mono	#ffffff	Mark on dark surfaces (canvas, images, embossing)
Black mono	#000000	Mark on light surfaces (light print paper, etching)

12.3 Forbidden Colors

- No neon cyan (#00FFCC, #00F0FF, #00E5FF) — explicitly banned.
- No cyan/blue gradients or bright blue accents.
- No legacy colorways (teal/violet/warm retired at certification).
- No out-of-palette hex drift — verify hex values against the approved table before export.
- Historical design-brief recolor candidates (Neon Green #39FF14, Electric Cyan #00E5FF) are not certified and must not be used.

12.4 Print Color

Token	Hex	CMYK approx	Purpose
---	---	---	---
Mythic Emerald	#10b981	C 84 / M 0 / Y 55 / K 7	Spot-foil match; rich emerald
Deep Jade	#059669	C 88 / M 0 / Y 60 / K 18	Depth accents
Soft Mint	#34d399	C 68 / M 0 / Y 50 / K 0	Highlight (print rarely uses)
Abyssal Slate	#060c09	C 80 / M 55 / Y 70 / K 95	Dark canvases
Bone	#f3f4f6	C 3 / M 2 / Y 3 / K 0	Off-white text

CMYK values are process approximations. For exact brand color use a single **spot emerald** (HYDRA EMERALD) — PMS 16-6340-class green, confirm the chip with the printer — or emerald foil. Never mix the mark from CMYK tints of two inks.

13. Iconography

13.1 Favicon (16 px)

The smallest and most demanding surface. Certified gates:

Gate	Requirement	Measured
---	---	---

Gate	Requirement	Measured
16 px legibility	Three distinct head masses + visible center core; no merged blob	■ 42% ink; lobe ink 31/39/37
Optical centering	Mark optically centered in tile	■ bbox 85.4% × 78.1% (≥75%)
Contrast	Silhouette legible on light + dark browser UIs	■

13.2 App Icons

Generated per `11_EXPORT_PIPELINE.md`: 16–2048 px PNG/WEBP, maskable 512, Apple Touch 180, Android Chrome 192/512, macOS/Windows/Linux/Electron sets (see Chapter 26). All derive from the certified master — never upscale from a smaller copy.

13.3 Favicon Set (Web)

Asset	File	Size	Notes
---	---	---	---
Browser favicon (SVG)	<code>/hydra-mark.svg</code> (served)	vector	<code>%BASE_URL%hydra-mark.svg</code> in <code>index.html</code>
Browser favicon (ICO)	<code>favicon.ico</code>	16/24/32/48/64 (+256)	Optional legacy fallback
Safari mask icon	<code>safari-pinned-tab.svg</code>	vector	Monochrome silhouette; black fill
Apple touch icon	<code>apple-touch-icon.png</code>	180×180	Emerald fill, solid tile
Android Chrome	<code>android-chrome-192/512.png</code>	192 / 512	Emerald fill
PWA manifest	<code>manifest.webmanifest</code>	—	References 192/512 icons

13.4 OpenGraph / Social Card

- File: `public/hydra_logo.jpg` — 1024×1024 JPEG.
- Field: abyssal `#060c09`; mark emerald `#10b981`, centered, dominant (84.6% × 77.1% coverage).
- Injected via `metadata.json` → `src/hooks/useOpenGraph.ts`, resolved through Vite base path for GitHub Pages subpath deploys.
- Twitter card: `summary_large_image`.

14. Motion System

14.1 Animation Principles

- 1 **Identity before effect.** Every animation preserves the “three heads, one body” read.
- 2 **Settle, don’t loop.** Approved intros reveal and settle. The only approved continuous treatment is the low-key pulse (loading).
- 3 **External lighting only.** Glow, shadow, and bloom live in CSS utilities, never inside the SVG.
- 4 **Motion is opt-in by size.** Heavy effects are for display sizes (≥96 px); at ≤32 px restrict to opacity/glow.

- 5 **Reduced motion wins.** Every animation honors **prefers-reduced-motion: reduce**.

Hard rule: *continuous rotation (.hydra-mark-spin) is retired and forbidden — the 3-fold mark is rotationally self-similar, so spinning reads as a generic fan and re-opens the rotor defect closed at certification.*

14.2 Approved Effects

Effect	Duration	Easing	Notes
---	---	---	---
Stroke Reveal	900–1400 ms	cubic-bezier(0.65, 0, 0.35, 1)	Heads then core; settle to full opacity fill
Outline Trace	700–1000 ms	ease-out	Outline variant only; may linger, then fill-in
Idle Animation	—	—	Static by default; no idle motion on the primary mark
Hover	150–250 ms	ease-out	Opacity/scale ≤ 1.05 , or glow on; no rotation
Glow Layer	300–600 ms	ease-in-out	.hydra-mark-glow ; follows currentColor
Pulse	3.2 s loop	ease-in-out	Loading only; opacity 1→0.55, scale 1→0.9
Rotation	—	—	FORBIDDEN
Loading Loop	single-shot or ≤ 3 repeats	—	Reveal + settle; never an infinite busy spinner

14.3 Technique Layers

- **SVG animation:** all paths carry **pathLength={1}** — **stroke-dasharray: 1;** **stroke-dashoffset: 1** → **0** per path reveals heads and core. Draw order: heads (0, 1, 2) first, then the core.
- **CSS animation:** **.hydra-mark-glow** (drop-shadow 0 0 6px / 0 0 16px **currentColor**) and **.hydra-mark-pulse**. New utilities live in **src/index.css** under the mark block and respect reduced motion. Prefer **opacity** / **filter** / small **scale**.
- **Framer Motion:** the site motion runtime. Animate the wrapper/motion element — **never** mutate **HydraLogo** path data or per-head transforms at runtime. Gate via **useReducedMotion()**.

14.4 Toolchain

Tool	Role	Rule
---	---	---
After Effects	Video / OG / marketing animations	Export from the master SVG; never re-trace; emerald family only
Lottie	Lightweight web animation	Via bodymovin from AE; honor Lottie.setReducedMotionHook
Rive	Interactive web states	Import master paths; single currentColor -equivalent palette
Blender	3D / depth treatments	Extrude/bevel only from the certified 2D silhouette

Cross-tool rule: every motion asset starts from `hydra-mark.svg` (or the identical `HydraLogo.tsx` geometry); exported vectors are re-imported and diffed for drift before release.

14.5 Timing & Rhythm

- Micro-interactions: 140–240 ms; reveals: 600–1400 ms; pulse loop: 3.2 s.
- Stagger: 60–90 ms offsets; the mark reveals before its supporting text.
- Loading sequences keep a total budget under **1600 ms**.

14.6 Restrictions

- 1
- No continuous rotation. 2. No per-head independent motion. 3. No color-shifting flashes. 4. No morphing paths. 5. No baked lighting. 6. No busy looping. 7. Reduced motion always honored.

15. Website Integration

15.1 Surface Map

Surface	Component / Location	Implementation	Color
---	---	---	---
Navbar	<code>src/components/Navigation.tsx</code>	HydraLogo	Emerald #10b981 via text-accent
Hero	<code>src/components/Hero.tsx</code>	HydraLogo	Emerald #10b981
Footer	Footer (existing layout)	HydraLogo (or favicon raster fallback)	Muted emerald / white mono
Loading	<code>src/components/LoadingSplash.tsx</code>	HydraLogo + wordmark + emerald progress bar	Emerald
Splash	<code>src/components/PreSplashSelector.tsx</code>	HydraLogo selection tile	Emerald
Watermark	Full-page / section watermark	HydraLogo at ≈8–14% opacity	Low-opacity mono
OpenGraph	<code>public/hydra_logo.jpg</code> via <code>useOpenGraph.ts</code>	1024×1024 raster	Emerald on Abyssal
Manifest icons	<code>/manifest</code> icons	192 + 512 px	Emerald fill
Safari mask icon	<code>safari-pinned-tab.svg</code>	Monochrome silhouette	Black fill
Dark mode	<code>html.dark</code> + <code>localStorage['hydra-theme']</code>	Emerald on Abyssal #060c09	Emerald
Light mode	default canvas #f8faf9	Emerald on light canvas	Emerald (black mono in print)

15.2 Navbar

- Mark size 24–28 px rendered height (never below 24 px).

- Color **#10b981**, inherited via **text-current**.
- Lockup: mark + wordmark (typography only). No text inside the mark SVG.
- Clear space: one hex-core radius (17 u ≈ one head-height) on every side.
- Hover: optional **.hydra-mark-glow**; no scale beyond 1.05×; no rotation.

15.3 Hero

- Mark size display scale, 96–160 px rendered height.
- Placement: optically centered relative to the headline, shared optical axis.
- Motion: static by default; pulse permitted on loading only; rotation forbidden.
- Reduced motion must freeze all mark animation.

15.4 Loading & Splash

- LoadingSplash: **HydraLogo** + "HYDRA SAMO" wordmark + a thin emerald gradient progress bar (≈1.9 s) with a live **{percent}%** counter — the counter is the only text under the bar (no status/progress caption).
- The splash surface is **transparent**: the fixed **AmbientBackground** shows through, so the abyssal emerald glow reads as one living backdrop from first paint.
- The mark gently pulses via a motion wrapper (opacity 0.15→1→0.15, 2.6 s loop, **easeInOut**); clicking skips; reduced-motion users jump straight to content.
- PreSplashSelector: mark as selectable identity tile; emerald fill, glassmorphic tile.
- Approved loading treatment is the **pulse**, never a spin; a stroke-draw reveal may accompany intros but must complete and settle.

15.5 Watermark

8–14% opacity of **currentColor** (never above 20%); white or emerald mono; up to 60–80% of section height; static, slow fade-in on scroll permitted.

15.6 Dark / Light Mode

Token	Dark (html.dark)	Light
---	---	---
Canvas	#060c09 (Abyssal Slate)	#f8faf9
Mark	Emerald #10b981	Emerald #10b981
Monochrome fallback	White mono for overlays	Black mono for print/light-critical
Glass layer	rgba(255, 255, 255, 0.04) + backdrop-blur-md	keep translucent

16. Development Architecture

16.1 Stack

- **React + Vite** (TypeScript), Tailwind v4 CSS-first config in **src/index.css** (**@import "tailwindcss"**).

- Dark mode is class-based (`html.dark`), controlled by `App.tsx` and `localStorage[ 'hydra-theme' ]`.
- Class logic always through `cn()` from `src/lib/utils.ts`.
- **Relative imports only** (`../components/...`) — the `@` alias points to root, never `src/`.
- Motion: Framer Motion + Lenis smooth scroll.
- Package name: `react-example`.

16.2 Scripts

Command	Action
---	---
<code>npm run dev</code>	Dev server on port 3000 (host 0.0.0.0), not Vite's default 5173
<code>npm run lint</code>	<code>tsc --noEmit</code>
<code>npm run build</code>	Production build
<code>npm run clean</code>	Removes <code>dist/</code>

16.3 Content Architecture

- All content from `src/data/portfolioData.ts`.
- `PROJECTS` is deliberately empty — `WorkGallery` must retain its empty-state handler.
- `VOICE_TRACKS` playback relies on `audioUrl` in `/public/audio/`.
- `DISCIPLINES` IDs are strictly `'video' | 'motion' | 'voice'`.
- Metadata handled client-side via `src/hooks/useOpenGraph.ts` pointing to `/hydra_logo.jpg`.
- Contact form uses `VITE_FORM_ENDPOINT` / `VITE_FORM_ACCESS_KEY`; unconfigured warning state is kept.
- UI glyphs come from `lucide-react` — stock components only (`Play`, `Pause`, `Volume2`, `VolumeX`, `Mic`, `MicOff`, `FileText`, `Film`, `Layers`, `X`, `Check`, `CheckCircle2`, `ArrowRight`); no hand-rolled SVG glyph sets.
- `VoiceOverSection` renders a "coming soon" placeholder (no audio-player content), mirroring the `WorkGallery` empty-state pattern.

16.4 Device-Tier Performance Budget

Source of truth: `src/hooks/useDeviceTier.ts` detects `low | medium | high` once per session (CPU cores, deviceMemory, effectiveType, saveData, coarse pointer) and mirrors it onto `<html data-quality="low|medium|high">`.

Tier	Experience
---	---
Low	No Lenis (native scroll — anchors need <code>[id] { scroll-margin-top: 96px }</code>), 1 static ambient blob, no aurora/spotlight/drift, no backdrop-blur on glass, no blur entrances, every other waveform bar
Medium	Lenis at 0.95s, 2 ambient blobs, reduced <code>blur(9px)</code> glass, no spotlight
High	Full experience — Lenis 1.2s, 3 blobs + drift + aurora + spotlight, <code>blur(12px)</code> glass

- JS animation gated with `cheapMotion = isCoarse || tier === 'low'`; CSS compositing gated via `html[data-quality]` selectors.

16.5 Content Security Policy

`index.html` ships a CSP meta tag with `no `unsafe-eval`` — required for Electron to clear its "Insecure Content-Security-Policy" warning. `script-src 'unsafe-inline'` is kept for Vite dev; do not add `unsafe-eval` back.

17. SVG Engineering

17.1 The Master File

`public/hydra-mark.svg` — the canonical static SVG (favicon, downloads, static tools), generated from the same geometry as `src/components/HydraLogo.tsx`:

```
<font face="Courier" size="7.4" color="#d6e2e8">&lt;svg xmlns="http://www.w3.org/2000/svg" viewBox="11 2.94 78 78"&gt;
<font face="Courier" size="7.4" color="#d6e2e8">  &lt;path fill="#10b981" d="M 50.00 11.50 C ... Z" /&gt;</font>
<font face="Courier" size="7.4" color="#d6e2e8">  &lt;path fill="#10b981" transform="rotate(120 50 50)" d="M 50.00 11
<font face="Courier" size="7.4" color="#d6e2e8">  &lt;path fill="#10b981" transform="rotate(240 50 50)" d="M 50.00 11
<font face="Courier" size="7.4" color="#d6e2e8">  &lt;path fill="#10b981" d="M 50.00 33.00 L ... Z" /&gt;</font>
<font face="Courier" size="7.4" color="#d6e2e8">&lt;/svg&gt;</font>
```

17.2 Engineering Standards

Standard	Value
---	---
Geometry	Certified V4; identical across all variants (single unique head-path hash)
Frame	<code>viewBox="11 2.94 78 78"</code> on all production variants
Paths	Exactly 4; never merged; only the certified <code>rotate(120/240 50 50)</code> transforms
Color	Pure <code>currentColor</code> fills/strokes — recolorable to any certified colorway with no geometry change
Decoration	No gradients, glow, shadows, reflections, or <code><defs></code> lighting in the SVG
Motion-readiness	<code>pathLength={1}</code> on every path for draw/trace math
Accessibility	<code>aria-hidden="true"</code> , <code>focusable="false"</code> , <code>data-hydra-mark</code>
Optimization	Coordinate rounding / markup hygiene permitted only with zero visual change

17.3 Single Source of Truth

- `src/components/HydraLogo.tsx` — live geometry used by every UI surface.

- **public/hydra-mark.svg** — canonical static SVG (favicon, downloads, static tools), generated from the same geometry.
- **public/hydra_logo.jpg** — derived OpenGraph / Twitter card (1024×1024).

***Rule:** every raster, icon, social asset, print file, or motion asset MUST be generated from the certified geometry. It is forbidden to re-draw, re-trace, or re-imagine the mark for any export. Re-export only, never re-create.*

17.4 Integrity

- Master SHA-256
f12de4fdd94c2dbd155b0c4a23c80cdbb6420bd394a601492ade20d23380c1fa.
- Any file whose checksum diverges from the recorded value on a released version is NOT the certified master and must be quarantined.
- Verified: fresh rsvg render of the master is byte-identical to the shipped
PNG/hydra-mark-emerald-256px.png.

18. Component Implementation

18.1 HydraLogo.tsx

```

<font face="Courier" size="7.4" color="#d6e2e8">import React from 'react';</font>
<font face="Courier" size="7.4" color="#d6e2e8">import { cn } from '../lib/utills';</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">interface HydraLogoProps {</font>
<font face="Courier" size="7.4" color="#d6e2e8">  className?: string;</font>
<font face="Courier" size="7.4" color="#d6e2e8">  variant?: 'fill' | 'outline';</font>
<font face="Courier" size="7.4" color="#d6e2e8">}</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">const HEAD_PATH =</font>
<font face="Courier" size="7.4" color="#d6e2e8">  'M 50.00 11.50 C 50.56 12.05, 52.20 13.55, 53.36 14.80 ... Z';</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">const CORE_PATH =</font>
<font face="Courier" size="7.4" color="#d6e2e8">  'M 50.00 33.00 L 64.72 41.50 L 64.72 58.50 L 50.00 67.00 L 35.28 58.50 ... Z';</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">export const HydraLogo: React.FC<HydraLogoProps> = ({</font>
<font face="Courier" size="7.4" color="#d6e2e8">  className,</font>
<font face="Courier" size="7.4" color="#d6e2e8">  variant = 'fill',</font>
<font face="Courier" size="7.4" color="#d6e2e8">}) => {</font>
<font face="Courier" size="7.4" color="#d6e2e8">  const isOutline = variant === 'outline';</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">  return (</font>
<font face="Courier" size="7.4" color="#d6e2e8">    <svg</font>
<font face="Courier" size="7.4" color="#d6e2e8">      viewBox="11 2.94 78 78"</font>
<font face="Courier" size="7.4" color="#d6e2e8">      data-hydra-mark</font>
<font face="Courier" size="7.4" color="#d6e2e8">      aria-hidden="true"</font>
<font face="Courier" size="7.4" color="#d6e2e8">      focusable="false"</font>
<font face="Courier" size="7.4" color="#d6e2e8">      className={cn('hydra-mark block text-current', className)}</font>
<font face="Courier" size="7.4" color="#d6e2e8">    <g></font>
<font face="Courier" size="7.4" color="#d6e2e8">      <g</font>
<font face="Courier" size="7.4" color="#d6e2e8">        className="hydra-heads"</font>
<font face="Courier" size="7.4" color="#d6e2e8">        fill={isOutline ? 'none' : 'currentColor'}</font>
<font face="Courier" size="7.4" color="#d6e2e8">        stroke={isOutline ? 'currentColor' : 'none'}</font>
<font face="Courier" size="7.4" color="#d6e2e8">        strokeWidth={isOutline ? 3 : undefined}</font>
<font face="Courier" size="7.4" color="#d6e2e8">        strokeLinecap="round"</font>
<font face="Courier" size="7.4" color="#d6e2e8">        strokeLinejoin="round"</font>
<font face="Courier" size="7.4" color="#d6e2e8">      <g></font>
<font face="Courier" size="7.4" color="#d6e2e8">        <path className="hydra-head hydra-head-0" d={HEAD_PATH} pa</font>
<font face="Courier" size="7.4" color="#d6e2e8">        <path className="hydra-head hydra-head-1" d={HEAD_PATH}</font>
<font face="Courier" size="7.4" color="#d6e2e8">          transform="rotate(120 50 50)" pathLength={1} /></font>
<font face="Courier" size="7.4" color="#d6e2e8">        <path className="hydra-head hydra-head-2" d={HEAD_PATH}</font>
<font face="Courier" size="7.4" color="#d6e2e8">          transform="rotate(240 50 50)" pathLength={1} /></font>
<font face="Courier" size="7.4" color="#d6e2e8">        <path className="hydra-core" d={CORE_PATH} pathLength={1}</font>
<font face="Courier" size="7.4" color="#d6e2e8">      </g></font>
<font face="Courier" size="7.4" color="#d6e2e8">    </svg></font>
<font face="Courier" size="7.4" color="#d6e2e8">  )</font>
<font face="Courier" size="7.4" color="#d6e2e8">}</font>

```

18.2 Usage

```

<font face="Courier" size="7.4" color="#d6e2e8"><HydraLogo className="h-7 w-7 text-accent" /></font>
<font face="Courier" size="7.4" color="#d6e2e8"><HydraLogo className="h-16 w-16 sm:h-20 sm:w-20 text-accent/70 hyd</font>
<font face="Courier" size="7.4" color="#d6e2e8"><HydraLogo className="h-12 w-12 sm:h-14 sm:w-14 text-[#f3f4f6] hyd</font>
<font face="Courier" size="7.4" color="#d6e2e8"><HydraLogo variant="outline" className="h-10 w-10 text-accent" /></font>

```

18.3 CSS Utilities

```
<font face="Courier" size="7.4" color="#d6e2e8">.hydra-mark-glow {</font>
<font face="Courier" size="7.4" color="#d6e2e8">  filter: drop-shadow(0 0 6px currentColor) drop-shadow(0 0 16px curr
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">.hydra-mark-pulse {</font>
<font face="Courier" size="7.4" color="#d6e2e8">  animation: hydra-pulse 3.2s ease-in-out infinite;</font>
<font face="Courier" size="7.4" color="#d6e2e8">}</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">@media (prefers-reduced-motion: reduce) {</font>
<font face="Courier" size="7.4" color="#d6e2e8">  .hydra-mark-pulse {</font>
<font face="Courier" size="7.4" color="#d6e2e8">    animation: none;</font>
<font face="Courier" size="7.4" color="#d6e2e8">  }</font>
<font face="Courier" size="7.4" color="#d6e2e8">}</font>
```

18.4 Component Rules

- Always pass class strings through `cn()`.
- The component renders decorative-only (**aria-hidden**); visible brand text carries meaning.
- Animate the wrapper, never the path data or per-head transforms.
- No baked gradients/glow in the component — all lighting via CSS utilities.

19. Asset Pipeline

19.1 Export Matrix

Format	Ext	Use	Resolution/Vector	Transparency
---	---	---	---	---
SVG	.svg	Master, web, favicon, vector editing, motion	Vector	Yes (native)
PNG	.png	Raster web assets, favicons, social, print raster	16–2048 px	Yes
WEBP	.webp	Compressed web delivery	512–2048 px	Yes
ICO	.ico	Windows browser favicon	Multi-res (16–64 px, +256)	Yes
PDF	.pdf	Vector print / client deliverables	Vector	Yes
EPS	.eps	Legacy vector print (Illustrator)	Vector	Yes
AI	.ai	Editable Illustrator master for print partners	Vector	Yes
FIG	.fig	Figma brand library / site UI token	Vector	Yes

19.2 Required Raster Sizes

16 (favicon minimum) · 24 (UI inline) · 32 (favicon/window) · 48 (app icon grid) · 64 (browser icon) · 128 (standard avatar) · 256 (app icon/storefront) · 512 (PWA/maskable) · 1024 (OG/social, App Store) · 2048 (print raster, retina canvas).

19.3 Naming Convention

```
<font face="Courier" size="7.4" color="#d6e2e8">hydra-{asset}-{colorway}-{variant}-{size}px.{ext}</font>
```

Segment	Allowed values
---	---
asset	mark, favicon, og, avatar, appicon, mask, touch, logo
colorway	emerald, jade, mint, white, black
variant	fill, outline
size	16 ... 2048 (vector formats omit)
ext	svg, png, webp, ico, pdf, eps, ai, fig

Examples: **hydra-mark-emerald-fill-512px.png** · **hydra-favicon-emerald-fill-32px.ico** · **hydra-og-emerald-fill-1024px.webp** · **hydra-mark-emerald-fill-master.pdf**.

Rules: lowercase, kebab-case, no spaces; version tag only on MASTER assets; never include author initials, dates, or "final/final2".

19.4 Variant Specification

- **Transparent:** fully transparent canvas, no matte baked in.
- **Monochrome:** **white_fill** (**#ffffff**) for dark surfaces; **black_fill** (**#000000**) for light surfaces. The mark is monochrome-true.
- **Outline:** emerald stroke **#10b981**, width 3 units at source scale, **stroke-linecap="round"**, transparent core, **no interior tail strokes**. For technical/annotation contexts only — not a substitute for the fill mark.

19.5 Quality Gates

Gate	Requirement
---	---
Source	Generated from master only; no re-drawing
Geometry	Unchanged — diff the SVG path data before/after export
Colors	Match approved palette (no hex drift)
Transparency	Alpha correct; no halo/background leak
Legibility	16 px shows three head masses + visible core (42% ink gate)
Naming	Conforms to the template
Checksum	Master checksum recorded in the release manifest
QA	Each batch passes the relevant items of Chapter 21

19.6 Retry / Re-export Policy

Re-exports are free and encouraged; **re-creations are prohibited**. If a tool introduces a rendering artifact, correct the tooling, never the mark.

20. Production Workflow

20.1 The Production Pipeline

01	MASTER ASSET	02	PRODUCTION QA	03	COMPLETION	04	RELEASE
	(12 packages,		(221/221 checks,		(158 assets +		(14 f
	147 files)		100/100 score)		SOURCE +		178 f
					UNSUPPORTED)		27 MB

20.2 Step 01 — Asset Production

12 packages, **147 files** in **ASSETS/**:

Package	Files
---	---
MASTER	8
SVG	5
PNG	17
FAVICON	15
WEB	23
SOCIAL	15
MOTION	9
PRINT	10
APP	29
SOURCE	9
DEVELOPMENT	7
README.md	1

Plus **UNSUPPORTED/** documenting proprietary formats. All generated from the certified master; geometry preserved exactly across all 13 master/production SVG variants (single unique head-path hash).

20.3 Step 02 — Production QA

All **147 files** inspected: XML + namespace + viewBox + path count for every SVG; dimensions, alpha, centering, clipping for every raster. **221 of 221 checks passed, score 100/100, zero warnings**. Details in Chapter 21.

20.4 Step 03 — Completion Audit

Resolved every fixable QA issue, generated every supported missing asset, archived legacy assets, documented unsupported proprietary formats (**.ai/.fig/.aep/.riv**) with READMEs rather than fabricating files. Repository became internally consistent. **STATUS: READY FOR RELEASE**.

20.5 Step 04 — Release Packaging

QA-passed tree packaged into **RELEASE/HYDRA_SAMO_BRAND_v1.0/** — **14 folders** (11 asset packages + DOCUMENTATION + LICENSE + REPORTS), **178 files**, enhanced **ASSET_MANIFEST.md**, release README, informational usage terms (no invented legal language), and archive **HYDRA_SAMO_BRAND_v1.0.zip** (27 MB, 205 entries, integrity test passed). Master checksum matches certified master byte-for-byte.

20.6 Step 05 — Final Sign-Off

Official project closure: **STATUS: COMPLETE** — the HYDRA SAMO Brand Identity System v1.0. Full lifecycle complete, visual identity officially frozen, repository approved for production deployment, archival, maintenance, and future brand applications.

21. QA Process

21.1 Validation Categories

The QA validation checklist (**OUTPUT/18_QA_VALIDATION.md**) defines PASS criteria, FAIL criteria, and verification methods across 12 categories:

- 1 **SVG Integrity** — geometry unchanged, class contract intact, no baked decoration, checksum matches.
- 2 **Favicon Readability** — 16 px legibility, optical centering, contrast.
- 3 **Website Integration** — one mark on all surfaces, favicon wired (**%BASE_URL%**), OG wired, build clean.
- 4 **Motion Compatibility** — stroke-draw readiness, no spin, motion settles, reduced motion.
- 5 **Print Readiness** — vector files, spot color named, minimum sizes, line widths.
- 6 **Responsive Scaling** — 16→2048 px legibility, retina, uniform aspect.
- 7 **Retina Displays** — 2× assets present, alpha preserved.
- 8 **Accessibility** — decorative mark, reduced motion, contrast.
- 9 **Animation Consistency** — one motion language, stagger order, color consistency.
- 10 **Folder Consistency** — hierarchy exists, MASTER immutable, ARCHIVE read-only.
- 11 **Naming Consistency** — convention followed, no version stamps in EXPORTS.
- 12 **Documentation Completeness** — all phase docs present, cross-references resolve, manifest up to date.

21.2 Result Recording

- **All PASS:** production-ready; proceed to Final Release.
- **Any FAIL:** block release; fix at the source, re-export, re-run the failed items, then re-run the full checklist.

21.3 Verified Result

The production QA run on 2026-08-05 audited **ASSETS/** (12 packages, 147 files) with automated scripts + manual spot checks (xmllint, ImageMagick, rsvg, sha256sum):

- **221 / 221 checks passed** — score **100 / 100**.
- Geometry preserved exactly across all 13 master/production SVG variants.

- All rasters derive from the certified vector; all motion respects the frozen motion system (no rotation).
- Favicon set: 16/32/48 PNG, apple-touch 180, android 192/512, mask-icon, `site.webmanifest`, `browserconfig.xml`, mstile 70/150/310 + wide 310×150 — all present, readable, centered.
- Print: vector PDF (v1.7) + EPS (DSC 3.0) + SVG confirmed vector; CMYK TIFF in CMYK colorspace; spot colors documented for vendor.
- App: 29 icons across Android, iOS, Windows, macOS, Linux, PWA, Electron, Chrome, Firefox, Safari.
- Unsupported proprietary formats (`.ai/.fig/.aep/.riv`) documented, not fabricated.
- No corrupted, missing, or misnamed assets.

22. Certification

22.1 Final Certification

Logo Version: V4 — refinement of V2: locked serpent-head geometry + R17 hex core, shared optical frame
`viewBox="11 2.94 78 78"`.

Field	Value
---	---
Certification date	2026-08-05
Overall score	59 / 100 (V2 consensus baseline, 4-reviewer pipeline)
Final verdict	■ CERTIFIED
Review cycle	First Review → REFINE → V4 Implementation → Final Validation
Basis	Full closure of all review findings by measured V4 verification; all acceptance gates PASS

22.2 Certification Statement

The V4 refinement converted every review finding into a verified, measurable correction without abandoning the original identity:

- **Perceptual category corrected** — 57.5° wedge (target 55–60°), 0.0° tangency; serpent heads on one body, not a rotor.
- **Smallest surface repaired** — 16×16 favicon shows three distinct head masses with a visible core, optically centered at 85.4% × 78.1%.
- **Optical frame corrected** — reframed to `11 2.94 78 78`; ~40% larger in every fixed slot.
- **Construction validated** — tangent-continuous head path; R17 core anchors the form; outline variant renders with no interior leak.
- **Single-identity discipline restored** — OG asset rebuilt from emerald source; legacy colorways retired.
- **Engineering standards upheld** — `currentColor`, `pathLength`, unmerged paths, stable classes, external CSS lighting, `aria-hidden`; lint + build clean.

22.3 Production Readiness Checklist

Item	Status
---	---
Brand identity validated	■
Hidden Hydra concept preserved	■
Rotor interpretation eliminated	■ (57.5° wedge, spin retired)
Geometry finalized	■ (wedge 57.5°, tangency 0.0°)
SVG engineering approved	■
Website integration approved	■ (lint + build clean, call sites unchanged)
Animation compatibility approved	■ (pathLength, stroke-drawing ready; spin removed)
Favicon approved	■ (3 head masses + visible core at 16px)
Scalability approved	■ (16px through 512px+)
Commercial usage approved	■

23. Brand Freeze

23.1 Freeze Declaration

The HYDRA SAMO logo has entered **permanent production status**. Effective immediately, future visual redesigns are prohibited. The V4 mark as certified is the brand's identity on every surface and in every derived asset.

Locked and may not be altered:

- Head path geometry and the 57.5° wedge / 0.0° tangency construction.
- Hexagon core (circumradius 17, pointy-top orientation).
- Shared optical frame `viewBox="11 2.94 78 78"`.
- 3-fold rotational symmetry about `(50, 50)` and the `hydra-head-0/1/2` + `hydra-core` class contract.
- Monochrome-true, `currentColor` rendering with external CSS lighting only.
- The retired continuous-rotation cue (`.hydra-mark-spin`) — remains forbidden.

23.2 Permitted Modifications

- SVG optimization (path size, coordinate rounding, markup hygiene) with **no visual change**.
- Export optimization; asset generation derived from the single source.
- Additional file formats; technical bug fixes (rendering, accessibility, packaging).
- Documentation updates.

No geometry changes may be made unless a future official rebranding initiative is approved by the brand owner.

23.3 Freeze Record

- Logo version: **V4** · Freeze date: **2026-08-05** · Approved by: brand owner.
- Status: ■ **LOCKED** — **Permanent production identity**.
- This freeze supersedes all prior review-stage flexibility.

24. Governance

24.1 Clear Space

Keep an exclusion zone equal to **one hex-core unit** (17 units ≈ one head-height ≈ the mark’s inner radius) clear on all four sides. At 24 px mark height → 24 px clear space; at 100 px → 100 px. Applies to text, other logos, image edges, crop marks, decorative elements, and watermarks. Clear space is the minimum, never the target.

![[Spacing Diagram]](./ASSETS/SOURCE/hydra-mark_v4_spacing.svg)

24.2 Minimum Size

Surface	Minimum
---	---
Favicon / browser	16 px
UI (navbar, inline)	24 px
App icon / avatar	48 px
Social avatar	128 px (512 px source)
Print (ink)	12 mm mark height
Print (embroidery)	45 mm
Vector	no limit

Below the minimum, do not render the mark — use the wordmark/name in type instead.

24.3 Allowed Colors

Colorway	Hex	Where
---	---	---
Mythic Emerald	#10b981	Primary everywhere
Deep Jade	#059669	Secondary accents, depth
Soft Mint	#34d399	Glow / highlight treatments
White mono	#ffffff	Dark surfaces, imagery overlays, embossing
Black mono	#000000	Light surfaces, light print, etching

24.4 Background Rules

Background	Mark color
---	---
Abyssal #060c09 (default)	Emerald #10b981
Any dark image	Emerald with .hydra-mark-glow, or white mono

Background	Mark color
Light canvas / paper	Black mono (or emerald if contrast $\geq 4.5:1$ on that surface)
Mid-tone photography	White mono with soft drop shadow, or emerald in a glass chip
Anything busy	Keep the mark in a glass/abyssal tile

24.5 Incorrect Usage (never)

- 1
- Re-draw, re-trace, re-interpret, or simplify the mark.
- 2
- Stretch, squash, skew, or distort proportions.
- 3
- Rotate the mark (continuous or any non-certified fixed placement without approval).
- 4
- Add text, initials, slogan, borders, or badges inside/around the mark SVG.
- 5
- Apply gradients, glow, shadows, or reflections inside the SVG.
- 6
- Recolor to any forbidden color.
- 7
- Place on a clashing background or below minimum size.
- 8
- Animate with the retired spin treatment or any busy looping spinner.
- 9
- Mirror/flip the mark or break the 3-fold symmetry.
- 10
- Use a legacy raster or a screenshot as the brand asset.

24.6 Accessibility & Contrast

- All mark animation honors **prefers-reduced-motion: reduce**.
- The **HydraLogo** component renders **aria-hidden="true"** and **focusable="false"**; brand name lives in visible text.
- Emerald mark must maintain $\geq 3:1$ **contrast** against any background (WCAG AA for graphical objects); body text $\geq 4.5:1$, large text $\geq 3:1$.

24.7 Vector & Raster Rules

- Single source only; keep viewBox, **currentColor**, **pathLength**, unmerged paths, and the class contract.
- Generate rasters at 2× target (or 4× for ≤ 32 px) and downscale with LANCZOS; never upscale.
- Preserve alpha; no background matte in transparent exports.
- Store masters at the highest size (≥ 1024 px) and derive smaller sizes from the master raster.

24.8 Rotation & Distortion Policy

- **Continuous rotation: forbidden** (**.hydra-mark-spin** retired).
- **Static placement:** upright as certified; non-upright tilts limited to $\leq 15^\circ$ and must be approved by the brand owner per asset.
- **Distortion:** never stretched, squashed, reflected, or mirrored. Aspect ratio locked at 1:1; scale always uniform.

25. Repository Structure

25.1 Top-Level Repository

```
<font face="Courier" size="7.4" color="#d6e2e8">HYDRA SAMO LABS</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ src/ # Application source</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ components/ # HydraLogo, Navigation, Hero, WorkGal
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ data/portfolioData.ts # All content (single source)</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ hooks/ # useOpenGraph, useDeviceTier, ...</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ lib/utls.ts # cn()</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ index.css # Tailwind v4 + design tokens + mark u
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ public/ # hydra-mark.svg, hydra_logo.jpg, audi
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ ASSETS/ # Production asset packages (12 packag
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ OUTPUT/ # Complete documentation series (01-28
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ RELEASE/ # HYDRA_SAMO_BRAND_v1.0/ + release arc
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ REVIEWS/ # Review pipeline materials</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ DOCUMENTATION/ # This Brand Book (md/html/pdf + TOC)<
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ DESIGN.md # System design specification</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ AGENTS.md # Project + brand governance</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ BRAND_ASSET_PIPELINE.md # Pipeline overview</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ index.html # Shell, fonts, CSP, favicon link</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ package.json # Scripts + deps</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ vite.config.ts # Vite config (server block locked)</f
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ metadata.json # OG metadata source</font>
```

25.2 Repository ↔ Brand Drive Mapping

Repo path	Brand drive path
---	---
public/hydra-mark.svg	brand/MASTER/vector/hydra-mark_v4.svg
src/components/HydraLogo.tsx	brand/MASTER/component/HydraLogo.tsx
public/hydra_logo.jpg	brand/MASTER/social_card/hydra_logo_v4.jpg
OUTPUT/10_MASTER_ASSET.md ... 19_FINAL_RELEASE.md	brand/GUIDELINES/

26. Asset Library

26.1 Package Overview

ASSETS/ — 12 packages, 147 files, all QA-passed:

Package	Files	Contents
---	---	---
MASTER	8	hydra-mark_v4_*.svg (master, fill, outline, black, white, currentcolor, print, web) + README
SVG	5	Emerald/black/white/currentcolor fill + emerald outline
PNG	17	Emerald fill 16–2048 px (Lanczos-downscaled)
FAVICON	15	favicon.ico (4 sizes), favicon.svg, 16/32/48 PNG, apple-touch 180, android 192/512, mask-icon, manifest, browserconfig, mstile set
WEB	23	Navbar/hero/footer/loading/splash/watermark, OG + twitter (PNG & WEBP 1024×1024), favicon set

Package	Files	Contents
SOCIAL	15	Avatars / banners / covers per platform (e.g. YT banner 2560×1440)
MOTION	9	Stroke-reveal, outline-trace, loading-loop, idle-pulse, hover SVGs + spec/notes
PRINT	10	Vector PDF + EPS + SVG, CMYK TIFF, RGB/black/white/laser/outline 3000 px previews, README (spot colors to vendor)
APP	29	Android/iOS/Windows/macOS/Linux/PWA/Electron/Chrome/Firefox/Safari icons
SOURCE	9	Construction/spacing/palette/editable/currentcolor/web SVGs + export-spec, designer-notes, version-metadata, geometry-sheet
DEVELOPMENT	7	HydraLogo.tsx, cn helper, design tokens, tailwind tokens, usage examples, README, a11y notes
README.md	1	Package documentation

26.2 Unsupported Proprietary Formats

Format	Status
---	---
.ai	Documented — vector PDF/EPS/SVG shipped
.fig	Documented — SVG interchange
.aep	Documented — MOTION SVG + spec as reference
.riv	Documented — standard SVG geometry

Documented unsupported formats do not fail QA. Create the destination folder, generate README.md, explain why automatic generation is unavailable, and how to manually create the asset from the certified SVG.

26.3 Integrity

- Master SHA-256 **f12de4fd...1fa** byte-identical to **public/hydra-mark.svg**.
- Fresh rsvg render of the master is byte-identical to shipped **PNG/hydra-mark-emerald-256px.png**.

27. Print System

27.1 Print Surfaces

Surface	Key spec
---	---
Business cards	85×55 mm; mark ≥12 mm height; ≥300 gsm matte/soft-touch; spot-UV, black-on-black deboss, or emerald foil
Letterheads	A4; mark 18–22 mm; 0.4 pt hairline rule in Deep Jade or 20% emerald tint

Surface	Key spec
Invoices	A4; mark 16–18 mm; watermark ≤8% opacity
Contracts	A4; cover mark 28–32 mm; running footer 10 mm
Slides	16:9 (1920×1080); corner 24–32 px; title-slide center 96–128 px
Packaging	Mark ≥20 mm; never on a fold line or >3 mm from an edge
Merchandise	Chest 60–80 mm; sleeve 35–45 mm; mug 45–55 mm; die-cut stickers ≥25 mm

27.2 Specialty Processes

Process	Minimum	Key constraints
---	---	---
Embroidery	45 mm height	Min stitch 2 mm; one color max; throat notch needs ≥4 mm to read
Laser engraving	12 mm height	Min line 0.3 mm; filled silhouette best
Foil stamping	15 mm height	Min line 0.25 mm (engraved die); emerald foil preferred
Vinyl cutting	40 mm height	Min bridge 1.5 mm between islands

27.3 Minimum Line Widths

Offset/digital print 0.1 pt (rules 0.4 pt) · laser 0.3 mm · foil 0.25 mm · vinyl 1.5 mm bridge · embroidery 1.5–2 mm.

27.4 Color in Print

- Spot **HYDRA EMERALD** (PMS 16-6340-class green, confirm chip with printer) or emerald foil for logo-critical print.
- CMYK approximations in Chapter 12. Never mix the mark from CMYK tints of two inks.
- Vector files carry no fonts and no placed rasters unless intended; raster print ≥2048 px at 300 DPI final size.

28. Social System

28.1 Shared Rules

- Avatar:** mark alone — no text, initials, ring, or background pattern. Emerald fill on abyssal tile (or white mono on dark tile).
- Banner/cover:** emerald mark on abyssal field; brand name in typography only.
- Thumbnail:** mark ≥ 60% of the frame's shorter side.
- Safe area:** mark within center 60%; live text within center 80%.
- Cropping:** avatars crop hard to circles or rounded squares — leave ≥ one head-height margin between the mark's bounding box and the tile edge.

28.2 Platform Matrix

Platform	Avatar	Banner / Cover	Thumbnail / PFP	Safe area	Crop
---	---	---	---	---	---
Instagram	320×320	—	Post 1080×1080	center 60%	circle avatar
Behance	300×300	1600×400	1200×900	center 60%	avatar square
LinkedIn	400×400	1584×396	—	center 60%	circle PFP
GitHub	512×512	—	1280×640 social	center 60%	circle avatar
YouTube	800×800	2560×1440 (safe 1546×423)	1280×720	center safe area	circle PFP
Discord	512×512	960×540	128×128 emoji	center 60%	circle avatar
X (Twitter)	400×400	1500×500 (safe 1280×360)	1200×675 card	center 60%	circle photo
Portfolio	800×800	320×240	1024×1024 OG	center 60%	varies

28.3 Generation Notes

- Generate avatars at 512 px (or higher) from the master and downscale; never upscale.
- Avatar tiles square, mark optically centered at ≥75% tile coverage.
- One shared banner pattern — recreate per platform dimensions, never resize one banner across platforms.
- No location/geo badges anywhere in social assets.
- Platform crop fields change over time; re-verify before each release.

29. Motion Deliverables

ASSETS/MOTION/ (9 files) — all valid SVG with **pathLength="1"**, all honoring **prefers-reduced-motion**:

Asset	Spec
---	---
Stroke-reveal	Heads then core draw over 900–1400 ms; settles to full fill
Outline-trace	700–1000 ms; may linger as outline, then fill-in
Loading-loop	Pulse loop (3.2 s), single-shot or ≤3 repeats, then settle
Idle-pulse	Loading only; opacity 1→0.55, scale 1→0.9
Hover	150–250 ms; opacity/scale ≤1.05 or glow
Spec docs	rotation-forbidden.md , motion-spec.md , developer-motion-notes.md

Verified: no rotate/spin animation exists anywhere (keyframe scan); all effects honor reduced motion.

30. Website Assets

ASSETS/WEB/ (23 files) + the live site assets:

Asset	File	Status
---	---	---
Live geometry	src/components/HydraLogo.tsx	■ live
Standalone SVG	public/hydra-mark.svg	■ live
Social card	public/hydra_logo.jpg (1024×1024)	■ live
Favicon set	favicon.ico , apple-touch-icon.png , safari-pinned-tab.svg , manifest icons	■ generated
WEB package	Navbar/hero/footer/loading/splash/watermark + OG/twitter PNG & WEBP	■ generated

All generated from the master per the export pipeline and validated per QA. Favicon wired as **%BASE_URL%hydra-mark.svg**; OG resolved through Vite base path so GitHub Pages subpath deploys work.

31. Developer Guide

31.1 Getting Started

Command	Action
---	---
npm install	Install dependencies
npm run dev	Dev server on port 3000 (host 0.0.0.0)
npm run lint	TypeScript check (tsc --noEmit)
npm run build	Production build
npm run clean	Remove dist/

31.2 Conventions

- **Relative imports only** (**../components/...**) — never **@/components/**.
- Pass all class strings through **cn()** from **src/lib/utls.ts**.
- Use Tailwind v4 CSS-first config; dark mode via **html.dark + localStorage['hydra-theme']**.
- UI icons via **lucide-react** stock components — never hand-rolled replacement SVG glyphs.
- Do not modify the **server** block or **DISABLE_HMR** logic in **vite.config.ts**.
- Do not add **unsafe-eval** to the CSP in **index.html**.

31.3 Using the Mark

```
<font face="Courier" size="7.4" color="#d6e2e8">import { HydraLogo } from '../components/HydraLogo';</font>
<font face="Courier" size="7.4" color="#d6e2e8"></font>
<font face="Courier" size="7.4" color="#d6e2e8">&lt;HydraLogo className="h-7 w-7 text-accent" /&gt;</font>
<font face="Courier" size="7.4" color="#d6e2e8">&lt;HydraLogo variant="outline" className="h-10 w-10 text-accent" /&gt;</font>
```

- Fill is the default variant; outline is for technical/annotation contexts.
- All lighting via CSS utilities (`.hydra-mark-glow`, `.hydra-mark-pulse`) — never inside the SVG.
- Animation: reveal + settle; pulse for loading only; **no rotation**.

31.4 Environment Variables

- `VITE_FORM_ENDPOINT` / `VITE_FORM_ACCESS_KEY` — contact form (unconfigured warning state kept).
- `.env.example` documents all variables.

31.5 Content

- All content in `src/data/portfolioData.ts`. `PROJECTS` is deliberately empty — keep `WorkGallery`'s empty-state handler. Audio relies on `/public/audio/`. `DISCIPLINES` IDs: `'video'` | `'motion'` | `'voice'`.
- `VoiceOverSection` is a "coming soon" placeholder — do not restore the removed audio-player content unless approved.

32. File Structure

32.1 Production Hierarchy

The production archive layout under the brand asset root (`brand/` at the repository root or a managed drive):

```
<font face="Courier" size="7.4" color="#d6e2e8">brand/</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ MASTER/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ vector/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ component/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ social_card/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ MANIFEST.md
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ EXPORTS/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ svg/ png/ webp/ ico/ pdf/ eps/ ai/ fig/</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ (emerald_fill/ emerald_outline/ white_fill/ black_fill/</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ size folders 16/.../2048 as needed)</font>
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ WEBSITE/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ SOCIAL/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ PRINT/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ MOTION/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ GUIDELINES/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ ARCHIVE/
<font face="Courier" size="7.4" color="#d6e2e8">■■■■ RELEASES/

# Certified, immutable source assets +
# hydra-mark_v4*.svg (fill, outline, wh
# HydraLogo.tsx (live geometry mirror)</
# hydra_logo_v4.jpg (1024×1024 OG/Twitt
# checksums, owners, freeze record</fon
# Every generated format/size (disposab
# favicon/, manifest/, og/, ui/</font>
# instagram/ behance/ linkedin/ github/
# business-cards/ letterheads/ invoices
# packaging/ merchandise/ process-specs/
# ae/ lottie/ rive/ blender/ renders/</
# OUTPUT 10-19 production docs (mirror)
# v1/ v2/ retired-colorways/ (read-only
# v4.0.0/ (tagged release snapshots)</f
```

32.2 Directory Rules

- 1 **MASTER/** is immutable — writes require brand-owner approval.
- 2 **EXPORTS/** is disposable — regenerate any time; never hand-edit an export.
- 3 **ARCHIVE/** is read-only — retired assets kept for provenance, never reused.
- 4 **GUIDELINES/** mirrors `OUTPUT/` — the repo is the source of truth; the brand drive copy is a mirror.
- 5 **RELEASES/** holds a frozen snapshot per release tag so the brand can be reproduced from any release.

- 6
- Naming everywhere follows the export-pipeline template; no version stamps inside EXPORTS (the folder path carries it).

33. Version History

33.1 Brand Releases

Release	Date	Contents	Status
---	---	---	---
Brand v4.0.0 — "One Body, Three Heads"	2026-08-05	Production Asset Ecosystem — full pipeline	■ PRODUCTION-READY
Brand Identity System v1.0	2026-08-05	Repository closure + this Brand Book	■ COMPLETE
Brand Book v1.0 — maintenance r2	2026-08-05	Sync: LoadingSplash, lucide iconography, voice placeholder	■ COMPLETE
Brand Book v1.0 — maintenance r3	2026-08-06	Sync: repository optimization (STEP 28)	■ COMPLETE

33.2 Logo Versions

Version	Status
---	---
V1	Superseded (original concept)
V2	Superseded (pre-refinement baseline, 59/100)
V4	■ CERTIFIED — permanent production identity

33.3 Versioning Policy

Version	Meaning	Example
---	---	---
vX.0.0	Brand identity release (geometry/palette change — requires rebranding approval)	v5.0.0 — not anticipated
vX.Y.0	Asset-system release (new exports, formats, platforms)	v4.1.0
vX.Y.Z	Maintenance release (optimization, bug fixes, docs)	v4.0.1

The version numbering is independent of the logo **v{n}** label (the logo stays V4 unless a rebranding is approved). Every release updates the release doc, the asset manifest, and the checksum record.

33.4 Repository Optimization (STEP 28, 2026-08-06)

Engineering-only optimization per **OUTPUT/28_REPOSITORY_OPTIMIZATION.md** — "Optimize, Never Rewrite". No brand-visible change.

- **Removed:** **bun.lock** (npm-only project); **pdf-forge-exports/.../extracted-text.txt** (temporary artifact); empty gitignored **assets/** leftover (Phase 03 — **ASSETS/** remains the canonical brand archive).
- **Dependencies:** removed **motion** (duplicate of **framer-motion**), **autoprefixer**, **esbuild**, **tsx**; reclassified **vite**, **@tailwindcss/vite**, **@vitejs/plugin-react** to devDependencies. 0 vulnerabilities.
- **Dead code:** removed 47 unused translation keys × 3 locales + **localizeDisciplines/localizeVoiceTracks** helpers (~180 lines in **src/i18n/translations.ts**).
- **Performance:** main JS **495.88 kB** → **472.35 kB** raw (**154.81** → **150.15 kB** gzip); **VideoModal** (8.5 kB) and **ContactSection** (10.7 kB) lazy-loaded via **React.lazy** + **Suspense** (~19 kB off the critical path). Adaptive performance system (device tier → **html[data-quality]** + reduced-motion CSS) verified intact.
- **Accessibility:** state-aware localized **aria-label** on close/play/pause/mute controls and grade slider; **role="dialog"** + **aria-modal** on both modals; localized alt text on grade comparison images (5 new keys × 3 locales).
- **Identity (Phase 13):** repository migrated to **HYDRA SAMO/HYDRA SAMO LABS/** (workspace **HYDRA SAMO**); git remote already **github.com/hydra-samo/HYDRA-SAMO-LABS**; **MOTION_DIFF_REPORT.md** reference synchronized; this Brand Book's repo-structure diagram updated.
- **Untouched:** certified logo geometry, master SVGs, palette, typography, routes, **DESIGN.md**, **AGENTS.md**, governance docs, **RELEASE/** archive, **ASSETS/**.
- **Validation:** **npm run lint** ■, **npm run build** ■, production-preview smoke test ■ (all assets HTTP 200), responsive/asset audits clean.
- **Status:** **REPOSITORY STATUS: OPTIMIZED** — approved for long-term maintenance and production deployment.

34. Future Versioning

34.1 Future Work Scope

Future work is limited to:

- **Asset generation** — creating specified EXPORTS/, SOCIAL/, PRINT/, MOTION/ deliverables from the certified master.
- **Technical maintenance** — rendering/accessibility/packaging bug fixes; SVG optimization with zero visual change.
- **Export optimization** — new formats/sizes derived from the master.
- **Documentation updates** — keeping the production docs current.
- **Versioned releases** — tagging releases per the versioning policy.

Out of scope: visual redesign, geometry changes, palette changes, or reinterpretation. Any such request is a rebranding initiative requiring explicit brand-owner approval.

34.2 Version 1.1 Development

- Assigned to the Versioning Policy: **v1.1**, **v1.2** for asset-system additions; **v2.0** only via an approved redesign initiative.

- Repository is structured for version control, GitHub archival, future maintenance, professional client delivery, internal design-system usage, developer onboarding, and future website integration.

34.3 Archive Policy

- Every release snapshots its assets and manifest into ARCHIVE/ and RELEASES/.
- Superseded versions are read-only, preserved for provenance, never reused for production.
- The certified MASTER/ is never archived or overwritten — only versioned alongside new releases.

35. Final Release

35.1 Release Record

Field	Value
---	---
Brand release	v4.0.0 — "One Body, Three Heads"
Logo version	V4
Release name	Production Asset Ecosystem
System status	■ PRODUCTION-READY
Release date	2026-08-05
Certification	OUTPUT/08 — ■ CERTIFIED
Freeze	OUTPUT/09 — ■ LOCKED

35.2 Release Notes

- Phases 01–10 documentation complete (master asset, export pipeline, website assets, motion, social, print, governance, file structure, QA, release).
- Production pipeline executed end-to-end: 147 assets → 221/221 QA → 158-asset completion → 178-file release package → final sign-off.
- **What stayed untouched:** all SVG, source code, and the certified geometry. No redesign, no reinterpretation, no path modification.

35.3 Known Limitations

- 1 Print spot color chip must be confirmed with the print partner before the first print run.
- 2 Social platform dimensions change over time; re-verify against each platform before publishing.
- 3 Legacy history (V1/V2, retired colorways) exists only in ARCHIVE/ and git history — not production assets.

35.4 Release Package

RELEASE/HYDRA_SAMO_BRAND_v1.0/ — 14 folders, 178 files, plus archive
HYDRA_SAMO_BRAND_v1.0.zip (27 MB, 205 entries, integrity-tested). Master checksum matches certified master byte-for-byte. Release README documents brand overview, folder structure, per-discipline usage, and maintenance.

36. Appendix

36.1 Review Scores Detail

Category	Reviewer	V2 Baseline
---	---	---
Creative Direction	Creative Director	63 / 100
Construction	Senior Logo Designer	67 / 100
Brand Recognition	Brand Recognition Specialist	53 / 100
Risk / Gatekeeping	Devil's Advocate	54 / 100
Consensus Overall	All reviewers	59 / 100

Post-validation: all seven critical + five minor findings closed by measured V4 verification → ■ CERTIFIED.

36.2 Glossary

Term	Definition
---	---
Mark	The certified V4 SVG brand icon
Master	The single certified source geometry (hydra-mark.svg / HydraLogo.tsx)
Lockup	Mark + wordmark in typography (never inside the SVG)
Clear space	Exclusion zone of one hex-core unit per side
Core	The R17 pointy-top hexagon — the one body
Wedge	The 57.5° angular flare of each head at its base
Tangency	0.0° — tangent-continuous construction joints
Colorway	Approved palette member (emerald, jade, mint, white, black)
Spin	Retired continuous-rotation treatment — forbidden
Rebranding initiative	The only authorized path to geometry/palette change

36.3 Source Documents Index

Document	Role
---	---
DESIGN.md	System design specification
AGENTS.md	Project + brand governance
BRAND_ASSET_PIPELINE.md	Pipeline overview
OUTPUT/01-07	Review pipeline (reports, plan, changelog)

Document	Role
OUTPUT/08–09	Certification + freeze
OUTPUT/10–19	Production documentation series
OUTPUT/20–25 (+ reports)	Asset production, QA, completion, packaging, sign-off, execution
OUTPUT/26–27	Complete brand system + this book
ASSETS/SOURCE/geometry-sheet.md	Certified geometry record
ASSETS/DEVELOPMENT/	HydraLogo.tsx, tokens, usage examples, a11y notes

36.4 Master Checksum

```
<font face="Courier" size="7.4" color="#d6e2e8">SHA-256 (public/hydra-mark.svg)</font>
<font face="Courier" size="7.4" color="#d6e2e8">f12de4fdd94c2dbd155b0c4a23c80cdbb6420bd394a601492ade20d23380c1fa</font>
```

Final Declaration

STATUS: COMPLETE

This **HYDRA SAMO Brand Book (Version 1.0)** is the definitive reference manual for the HYDRA SAMO Brand Identity System.

This publication **supersedes the individual engineering documents for reading purposes**, providing one coherent, professional narrative of the complete brand lifecycle — while the original documents in the repository are **preserved as the official source files** and remain the authoritative records for governance and audit.

The HYDRA SAMO Brand Identity System is hereby declared:

- **Certified** — logo V4, permanently frozen.
- **Production-ready** — release v4.0.0 "One Body, Three Heads".
- **Complete** — full lifecycle closed 2026-08-05.

The brand is maintained through the semantic versioning policy; core logo geometry shall not be modified outside an approved rebranding initiative.

"One body, three heads. One engine, three crafts. Many heads, one unified vision."